Basic Code Analysis: Integrated Outage Troubleshooting Workshop

Duration : 90 Min.

Scenario

A major production crash occurred in the checkout cluster. Correlate Apache logs, Java Spring exceptions, database deadlocks, and configuration properties. Identify the root cause and draft a developer-ready post-mortem.

Target File Location & Creation

File to Create/Update: `student_workspace/solution.json` **Input Resource File to Inspect**: `student_workspace/outage_trace.log`

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Fill out 'post_mortem_report.json' in your workspace.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect `outage_trace.log` inside `student_workspace/`.
3. Open `solution.json` in `student_workspace/` and perform the required modifications.
4. Save your changes (`Ctrl + S` or `Cmd + S`).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your progress by running `python run.py` locally in the terminal.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Test Case	Requirement	Marks
TC1	post_mortem_report.json exists and is valid JSON	4 Marks

TC2	Root cause correctly identified as database connection pool exhaustion	4 Marks
TC3	Mismatched environment property config parameter mapped correctly	4 Marks
TC4	File name, class, and method of leak suspect correct	4 Marks
TC5	Outage timeline (start/end) correctly parsed from log lines	3 Marks
TC6	Business impact metrics correctly logged	2 Marks
TC7	Immediate recovery steps documented	2 Marks
TC8	Long-term architectural remediations included	2 Marks

Total Score: 25 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.